

Question 01)

What are AWS Global Infrastructure?

Answer

AWS Global Infrastructure is the worldwide network of data centers and networking resources that AWS uses to deliver cloud services globally.

It consists of three main components:

Regions: Geographic areas that contain multiple isolated data centers. Each Region is independent and provides low latency and fault tolerance.

Availability Zones (AZs): Physically separate data centers within a Region, connected by high-speed private networking. Using multiple AZs ensures high availability and reliability.

Edge Locations: Distributed locations worldwide used to cache and deliver content closer to users, reducing latency and improving performance.

Overall, AWS Global Infrastructure provides scalable, secure, highly available, and low-latency cloud services worldwide.

Question 02

What are the AWS infrastructure features?

Answer

AWS infrastructure provides a highly reliable and scalable cloud computing environment with the following key features:

Elasticity and Scalability: AWS can automatically scale resources up or down based on demand, ensuring efficient resource usage.

Fault Tolerance: AWS infrastructure is designed to continue operating even when a component fails, using redundancy and backup systems.

High Availability: Multiple Availability Zones and Regions ensure minimal downtime and continuous service operation.

Global Reach: AWS has data centers around the world, providing low latency and fast performance for global users.

Security: AWS provides secure infrastructure with physical protection, network security, and monitoring.

Overall, AWS infrastructure is designed to provide reliable, scalable, secure, and highly available cloud services.

Question 3

Write short notes on the AWS Region.

Answer

An AWS Region is a geographical area that contains multiple isolated Availability Zones. Each Region is independent and physically separated from other Regions.

Regions are designed to provide high availability, fault tolerance, and low latency. Customers choose a Region based on proximity to users, legal requirements, service availability, and cost.

Each Region has multiple Availability Zones connected by high-speed private networking to ensure reliability and continuous operation.

Question 4

How does AWS define Data Centers?

Answer

AWS defines Data Centers as secure physical facilities where AWS stores and processes customer data. These data centers contain servers, networking equipment, storage systems, and power infrastructure.

Each data center is designed with redundant power, cooling, and networking systems to ensure reliability. Data centers are grouped into Availability Zones, and each Availability Zone contains one or more data centers for fault isolation and high availability.

Question 5

Write the names of eight services under the AWS Management and Governance service category.

Answer

The following are eight AWS services under the Management and Governance category:

1. Amazon CloudWatch
2. AWS CloudFormation
3. AWS CloudTrail
4. AWS Config
5. AWS Trusted Advisor
6. AWS Organizations
7. AWS Systems Manager
8. AWS Control Tower

Question 6

Write the names of eight services under the AWS Management and Governance service category.

Answer

Eight services under AWS Management and Governance category are:

1. Amazon CloudWatch
2. AWS CloudTrail
3. AWS CloudFormation
4. AWS Config
5. AWS Trusted Advisor
6. AWS Organizations
7. AWS Systems Manager
8. AWS Control Tower

Question 7

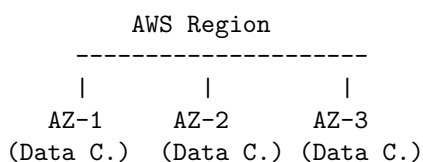
What do you know about the Availability Zone? Describe it with a figure.

Answer

An Availability Zone (AZ) is one or more physically separate data centers within an AWS Region. Each AZ has independent power, cooling, and networking infrastructure.

Availability Zones are connected by high-speed private networking. Deploying applications across multiple AZs ensures high availability and fault tolerance. If one AZ fails, the application continues running in another AZ.

Simple Figure Representation:



If AZ-1 fails -> AZ-2 or AZ-3 continues working

Question 8

What factors should you consider when selecting the correct region for your service in AWS?

Answer

The following factors should be considered when selecting an AWS Region:

- **Proximity to users:** To reduce latency and improve performance.
- **Compliance and legal requirements:** Data residency laws may require data to remain in a specific country.
- **Service availability:** Not all AWS services are available in every Region.
- **Cost:** Pricing may vary between Regions.

Question 9

Write the names of eight services under the AWS Management and Governance service category.

Answer

1. Amazon CloudWatch
2. AWS CloudTrail
3. AWS Config
4. AWS CloudFormation
5. AWS Trusted Advisor
6. AWS Systems Manager
7. AWS Organizations
8. AWS Service Catalog